

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
6	(a)	(i)	Reference to Faraday's law e.g. the emf is the rate of change/cutting of flux/flux linkage (1) The rate of change/cutting of flux (linkage) is proportional to the (angular) velocity (1)	1	1		2		
		(ii)	Agreement since straight line (through origin) or E proportional to A (1) Bad agreement at end (non-linear) (1) Drops off due to non-uniform field or equivalent (1)			3	3	1	
	(b)	(i)	Impedance of the inductor is zero/minimum when $f = 0$ (1) Short justification e.g. only resistance or $Z = R$ or hence $Z = \sqrt{X^2 + 82^2}$ is a minimum (1)		2		2	1	
		(ii)	$Z = \sqrt{82^2 + 82^2} = 116$ (1) $I = \frac{V}{Z}$ (1) Answer $I = 0.103$ [A] (1)	1	1		3	3	
	(c)	(i)	Phasors of capacitor and inductor are opposite direction or capacitor pd in antiphase of inductor (1) Cancel when $X_C = X_L$ or equivalent ($V_C = V_L$) (1) Gives minimum impedance / maximum current (1)	3			3		
		(ii)	Understanding impedance = resistance (1) $P = \frac{V^2}{R} = 1.76$ [W] (1)		2		2	1	
		(iii)	Substitution into $Z = \sqrt{(X_L - X_C)^2 + R^2} = 244$ [Ω] (1) Use of $I = \frac{V}{Z}$ (1) Answer = 49 m[A] (1)	1	1		3	3	
		(iv)	Calculation of Q factor (and realisation that this is germane) $Q = 1.9$ (1) Correct evaluation – not a particularly high Q factor or equivalent (can be obtained without Q factor but needs justification e.g. not sharp because resistance a little high (1)			2	2	1	
			Question 6 total	6	9	5	20	10	0